

Cycle-Time Optimization of Reconfigurable Robotic Pick-and-Place Cells in a Digital Twin

Abstract—Layout design for robotic pick-and-place cells is dominated by the cost of evaluating candidate configurations: each evaluation conventionally requires a sampling-based motion plan in a high-fidelity simulator, which is both slow and stochastic. We present a declarative reconfigurable-cell framework (ROS2 Jazzy + Gazebo Harmonic + MoveIt 2, UR5 + RG2 gripper) in which an optimizer searches over station poses and visit order using a deterministic joint-travel surrogate as its objective, rather than the simulator. The surrogate evaluates a configuration by chaining canonical, elbow-up inverse-kinematics solutions and summing joint-space travel, achieving $< 0.0003\%$ run-to-run spread. Our central contribution is an honest validation of this surrogate against real planned motion time. A naive comparison against a randomized feasibility planner (RRTConnect) yields no reliable correlation—the Pearson coefficient flips sign between repetitions ($+0.057$ vs. -0.550)—because per-plan path-length variance ($\sim 96\%$) swamps the between-configuration signal. Once the measurement is aligned with what the surrogate models—an optimizing planner (RRTstar), kinematic continuity (joint-to-joint goals), and a fixed visit order—the surrogate strongly predicts converged-optimal motion time: Pearson $r = 0.880$, Spearman $\rho = 0.943$ ($n = 6$). Optimizing against the surrogate with simulated annealing beats an equal-budget random baseline 5/5 (mean -17.4%), and the predicted reduction transfers to real motion: pooled $r = 0.876$, with optimized configurations running 19.4% faster in real RRTstar motion. Finally, an optimized configuration executes a full pick-place-pick-place relay end-to-end in the high-fidelity twin (two clean trials, no planning failures). We also report the boundary of the result: the most aggressive minimal-travel layouts are infeasible for the twin’s default randomized planner, so time prediction and full-twin executability are distinct guarantees.

Index Terms—robotic cell design, motion planning, digital twin, simulated annealing, surrogate optimization, reconfigurable manufacturing.

I. Introduction

The throughput of a robotic pick-and-place cell is largely fixed at design time by the relative placement of its work stations and the order in which the robot visits them. Searching this layout space is attractive but expensive: the natural objective—real cycle time—is obtained only by executing or planning each candidate in a simulator. Sampling-based planners such as RRTConnect [1] are fast but stochastic, so a single configuration must be evaluated many times to average out path-length variance, and a global search over hundreds of candidates becomes intractable.

A common remedy is a cheap surrogate objective. The risk is that the surrogate optimizes a proxy that does not track the quantity of interest. This paper takes the validation of that proxy as seriously as the optimization

itself. We build a declarative, reconfigurable cell on ROS2 and Gazebo (Section II), define a deterministic joint-travel surrogate (Section III), and then ask—repeatedly and honestly—whether it predicts real planned motion time (Section IV). We deliberately report two failed attempts before the successful one, because the failure mode is instructive: the surrogate is only predictive when the measurement is methodologically aligned with what it models. We then optimize against the validated surrogate and confirm that the predicted gains transfer both to real planned motion (Section V) and to full execution in the high-fidelity twin (Section VI).

Contributions. (i) A declarative reconfigurable-cell framework with a reusable headless IK oracle shared between synthesis and simulation. (ii) A deterministic joint-travel surrogate with $< 0.0003\%$ spread, suitable as a stable optimization objective. (iii) A rigorous surrogate-to-real validation establishing that the surrogate predicts an optimizing planner’s motion time ($r = 0.88$), together with a clear account of why it fails against a randomized planner. (iv) An end-to-end demonstration that simulated-annealing optimization against the surrogate yields configurations that are faster in real motion and executable in the twin, and an honest characterization of where executability breaks down.

II. System Architecture

The platform is a digital twin built on ROS2 Jazzy and Gazebo Harmonic, with a Universal Robots UR5 arm and an OnRobot RG2 two-finger gripper (Fig. 1). Motion planning uses MoveIt 2 with the OMPL planner suite. A cell is described declaratively in a locked schema—robot mount pose, belt geometry, part geometry, and a list of conveyor stations (each with an (x, y, ψ) pose), plus a relay task as an ordered list of pick/place operations. A generator realizes a configuration document into a `scene.yaml` and `task.yaml` consumed by the simulation.

Two components are shared, by import, between the headless optimizer and the live simulator, which is what makes the surrogate trustworthy: (1) an IK reachability guard that rejects any configuration with an unreachable station, and (2) the inverse-kinematics oracle used to score configurations. Grasping in the twin is implemented as a welded detachable joint, so the focus of this study is reach and transit motion rather than contact dynamics.

A practical but important lesson during twin construction: the `ros_gz_sim` spawn utility silently ignores a pose



Fig. 1. The high-fidelity Gazebo Harmonic twin for an optimized configuration (val_0): a UR5 + RG2 arm on a pedestal serves three conveyor stations against a warehouse backdrop. Station poses and visit order are produced by the optimizer; the green cube is the transferred part.

embedded in an SDF string, so every runtime fixture must be spawned via explicit `-x/-y/-z/-Y` flags; otherwise all fixtures stack at the world origin. Belt-support slabs are added so the part rests at a stable grasp height.

III. Deterministic Joint-Travel Surrogate

The surrogate estimates the motion cost of a configuration without invoking a planner. For each target pose in the (ordered) relay, it computes inverse kinematics and selects the canonical solution—the collision-free branch closest to a fixed elbow-up seed over a deterministic seed bank—then sums the joint-space distance traveled between consecutive canonical configurations. Selecting a canonical branch is essential: a generic IK call random-restarts and jumps between elbow-up/elbow-down solutions, which made an early version of the surrogate vary by 50–127% on identical inputs. With canonical selection the surrogate is effectively deterministic, with measured run-to-run spread below 0.0003%. This stability is a prerequisite for using it as an optimization objective: simulated annealing requires that two evaluations of the same point return the same value.

The visit order is fixed separately by a brute-force traveling-salesman solution over the stations, so that the surrogate and any downstream real measurement use an identical order.

IV. Surrogate-to-Real Validation

We treat “does the surrogate predict real motion time?” as a hard gate that must pass before any optimization result is trusted. The validation set val_hb contains six valid configurations spanning a range of surrogate values, all sharing one robot mount.

A. Two honest negatives

Attempt 1 (inconclusive). Measuring full cycle time in the twin mixed in ~ 130 s of configuration-independent reset/settle overhead and suffered $\sim 2\times$ run-to-run motion variance from the randomized planner; no clean trend emerged, and sim instability prevented collecting enough runs to average it out.

Attempt 2 (reliable negative). A headless probe stripped the fixed overhead and the wall-clock noise by summing planned trajectory duration with the real RRTConnect planner, averaged best-of-10. The result was explicitly not a relationship: the Pearson coefficient flipped sign between repetitions of the same experiment ($+0.057$ at 12 plans vs. -0.550 at 40 plans). The cause is that RRTConnect path duration retains $\sim 96\%$ per-plan spread even at best-of-10, and frequently fails on the collision-constrained reaches. A coefficient that flips sign is measuring noise; we report it as such rather than selecting the favorable run.

B. Aligning the measurement (Attempt 3)

The diagnosis was that minimal joint travel is the wrong target for a randomized, feasibility-only sampler. Three changes align the real measurement with the quantity the surrogate models:

- 1) Optimizing planner. RRTConnect $\rightarrow$ RRTstar [2], which is asymptotically optimal in path cost. This required injecting a group-qualified planner configuration so RRTstar is selectable for the arm group.
- 2) Kinematic continuity. The relay is planned joint-to-joint: each station’s goal is the collision-free IK branch closest to the previous station’s joint state, eliminating spurious elbow-flip excursions. A transfer $A \rightarrow B$ excludes both endpoint belts (leaving A and approaching B are required, not collisions), while every other belt remains an obstacle.
- 3) Fixed visit order. The brute-force TSP order is shared by surrogate and real measurement.

With continuity, five of six configurations become near-deterministic (per-plan spread $89\% \rightarrow 0\%$); the lone exception, val_2, is RRTstar occasionally failing to converge to its optimum.

C. Result

Table I reports the surrogate and the real RRTstar motion time (minimum, mean, median, std over 10 plans). Because RRTstar is asymptotically optimal, the minimum planned duration is the best estimate of its converged-optimal path cost; the mean is contaminated by non-converged runs (visibly val_2, whose min 7.35 s matches its mid-range surrogate while unlucky runs inflate its mean to 12.6 s). Table II compares correlation under each estimator; the min is both the most principled and the strongest. Fig. 2 shows the scatter and fit.

Once the measurement matches what the surrogate models, the deterministic joint-travel surrogate strongly predicts the optimizing planner’s motion time, and the

TABLE I

Surrogate vs. real RRTstar motion time on val_hb ($n = 6$, RRTstar $6s \times 10$ plans). Min is the converged-optimal estimate.

config	surrogate	min (s)	mean (s)	median (s)	std
val_0	5.17	6.81	6.81	6.81	0.00
val_1	6.38	8.06	9.14	8.92	0.98
val_2	6.60	7.35	12.58	10.40	3.79
val_4	6.92	8.56	8.61	8.61	0.02
val_3	6.96	9.03	9.08	9.10	0.03
val_5	7.36	9.11	9.52	9.11	1.22

TABLE II

Correlation of the surrogate with real motion time, by estimator.

estimator	Pearson r	Spearman ρ
min (converged-optimal)	0.880	0.943
median	0.704	0.543
mean	0.486	0.371

rank order is almost exact ($\rho = 0.943$). Attempt 2's negative is therefore a property of the randomized planner/sampler, not of the surrogate. The gate passes, justifying optimization against the surrogate.

V. Cycle-Time Optimization and Closing the Loop

A. Simulated annealing against the surrogate

We optimize station poses (x, y, ψ) and visit order with simulated annealing [3]. Moves comprise order swaps, global jumps, and local nudges, annealed with a cooling temperature and reheating; the search is seeded from the best of several random valid configurations. The fair baseline is the best random-valid configuration found under an identical oracle-evaluation budget, so any improvement is attributable to the search, not to extra evaluations. Across five specifications (5 seeds, 400 iterations, three stations), simulated annealing beats the equal-budget random baseline 5/5, with a mean improvement of 17.4% (Table III, Fig. 3).

B. The predicted reduction transfers to real motion

We then run the five optimized winners back through the headless RRTstar probe and pool them with the six prior valid configurations ($n = 11$). The correlation holds on min motion (Pearson $r = 0.876$, Spearman $\rho = 0.900$). All five optimized winners are perfectly deterministic over 10 plans (std 0.00) and sit at the low real-motion end. Their mean real motion is 6.79s versus 8.42s for the random valids—a 19.4% reduction that empirically matches the −17.4% surrogate gain (Table IV, Fig. 4). The loop is closed: a reduction in the cheap surrogate produces a reduction in real optimizing-planner motion of the same magnitude.

VI. High-Fidelity Twin Execution

To confirm that an optimized configuration is not merely fast on paper, we executed a feasible optimized winner

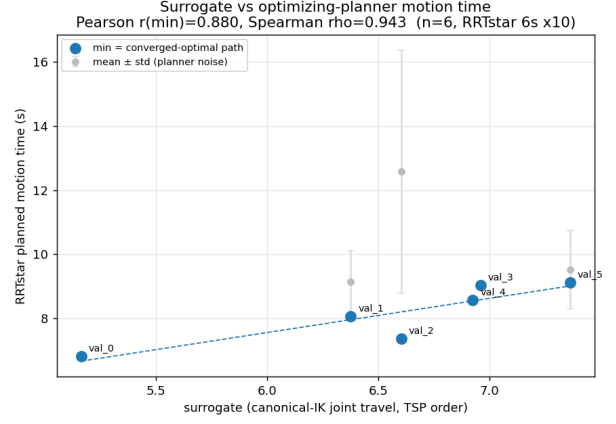


Fig. 2. Surrogate vs. real RRTstar motion time on val_hb. Filled points and fit line use the converged-optimal (min) estimate ($r = 0.880$, $\rho = 0.943$); faded points show mean $\pm$ std for transparency.

TABLE III

Simulated annealing vs. equal-budget random baseline (3 stations, 400 iters).

seed	SA cost	baseline	impr. vs. base	best order
100	4.750	6.491	26.8%	c3 c1 c2
101	4.453	6.154	27.7%	c3 c1 c2
102	4.468	5.210	14.2%	c2 c1 c3
103	4.774	5.355	10.8%	c1 c3 c2
104	4.887	5.287	7.6%	c1 c3 c2
mean	4.667	5.699	17.4%	—

(val_0, surrogate 5.19) in the full high-fidelity twin: Gazebo Harmonic GUI, the complete move_group stack, the real RG2 gripper, and a welded-cube carry. The IK guard passed (all three stations reachable), and the four-operation relay (pick-place-pick-place) ran end-to-end across a warmup and a measured trial, both succeeding with the gripper verified clear at start and no planning failures (Table V, Fig. 5).

VII. Discussion and Limitations

The result is bounded in an important way. The most aggressive minimal-travel winners produced by the optimizer place stations close together; while the headless RRTstar probe (with belt exclusion and joint continuity) solves these, the twin's default randomized planner (RRT-Connect with the real gripper collision model and a welded carry) cannot execute the cramped later segments—the first transfer succeeds, then a subsequent segment fails to plan. Thus our guarantees separate cleanly: the surrogate is validated as a predictor of optimizing-planner motion time, and an optimized configuration is demonstrated executable in the twin, but the single most aggressive optimum is not necessarily both. A natural extension is to fold a twin-executability constraint (e.g., a minimum inter-station clearance, or planning with the same planner the twin uses) directly into the surrogate or as an

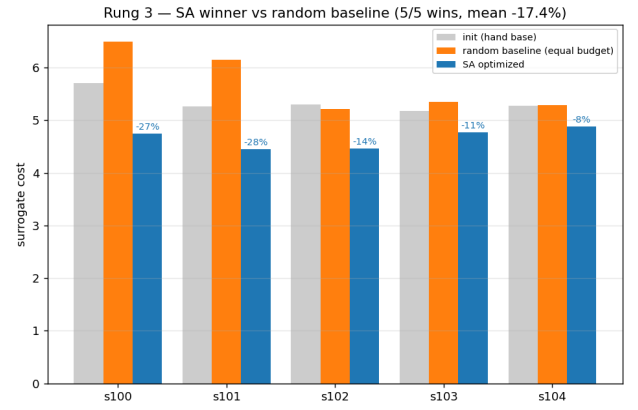
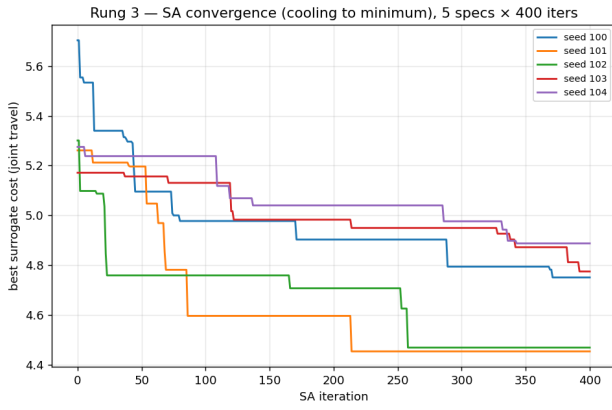


Fig. 3. Simulated annealing against the surrogate. Left: convergence trajectories (surrogate cost vs. iteration) for the five seeds, cooling to a global minimum. Right: final optimized cost vs. the equal-budget random baseline per seed; SA wins 5/5 (mean -17.4%).

TABLE IV

Closed loop: optimized winners vs. random valids (pooled $n = 11$, min motion).

group	mean real motion (s)	pooled correlation
SA winners ($n = 5$)	6.79	$r = 0.876, \rho = 0.90$
random valids ($n = 5$)	8.42	
relative improvement: -19.4% real motion (-17.4% surrogate)		

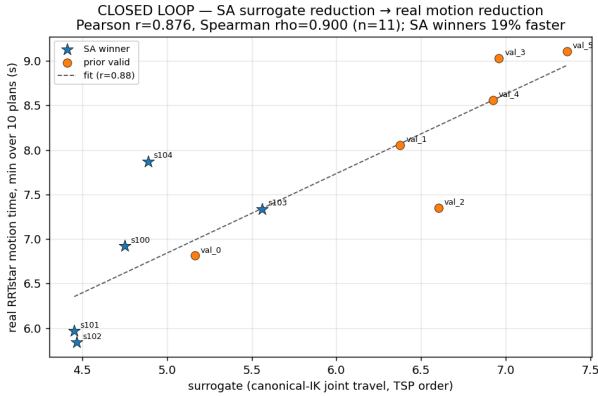


Fig. 4. Pooled surrogate vs. real motion ($n = 11$). Optimized winners (low end) and random valids lie on the same trend; the surrogate reduction maps onto a real-motion reduction.

optimization constraint. Other limitations include welded grasping (contact dynamics are out of scope) and a three-station relay; scaling the station count and the part set is future work.

VIII. Conclusion

We presented a reconfigurable robotic pick-and-place cell whose layout is optimized against a deterministic joint-travel surrogate, and validated that surrogate against a high-fidelity twin with explicit attention to honesty: two documented negative results precede the positive one, and the boundary of the positive result is stated. When

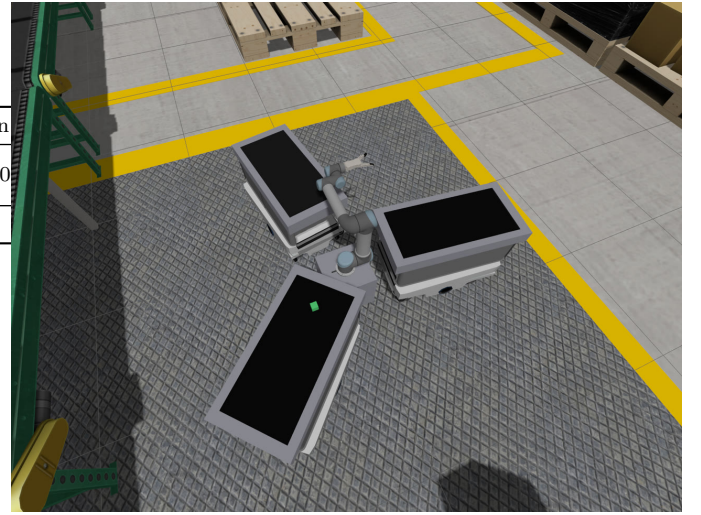


Fig. 5. Twin execution of the optimized winner val_0: the UR5 + RG2 arm reaches into a conveyor station mid-relay. The three stations are pinwheelled around the arm in the optimizer’s chosen layout; the welded part (green cube) is carried between stations. Both trials complete the full pick–place–pick–place relay with no planning failures (Table V).

TABLE V

Full-twin relay execution of optimized winner val_0.

run	cycle (s)	success	gripper clear	failure
warmup	133.0	yes	yes	none
trial 1	148.0	yes	yes	none

the real measurement is aligned with what the surrogate models—an optimizing planner, kinematic continuity, and a fixed visit order—the surrogate predicts converged-optimal motion time ($r = 0.88$, $\rho = 0.94$). Simulated annealing against the surrogate beats an equal-budget baseline 5/5 (-17.4%), and the gain transfers to real motion (-19.4%) and to a clean end-to-end relay in the twin. The methodology—validate the proxy as rigorously

as the optimization, and report where it stops holding—is the central takeaway.

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